

What is Swarm? tl;dr

UNSTOPPABLE STORAGE

Swarm is a **decentralised storage and communication network** run by peer-to-peer **Bee nodes**. Upload data, get back a **content hash** — a permanent address derived from the bytes themselves. Anyone with the hash can retrieve the content from any Bee node or public gateway.

Storage is bought as **postage stamps**: prepaid batches purchased with **xBZZ** on Gnosis Chain. Batches have a TTL and expire unless topped up. If every batch covering a chunk expires, that content can be pruned.

Homepage



Swarm
ethswarm.org

Docs



Documentation
docs.ethswarm.org

Explorer



SwarmScan
swarmscan.io

What you can build on Swarm

BUILD WITH FREEDOM · NO SERVERS, NO GATEKEEPERS

File & media storage

Fixed, content-addressed URLs for any file — images, video, PDFs, datasets, ML artifacts, NFT assets.

Live video & streaming

Early proof-of-concept work on HLS over Swarm exists — expect to integrate and tune it yourself.

Decentralised websites

Publish a static front-end straight to the network. No server, no CDN, no DNS lock-in.

Chat, comments & messaging

Real-time chat, threaded comments and peer messaging. Real-time delivery needs a GSOC subscription (see back).

Blogs & dashboards

Stable URLs whose contents update via feeds. Personal blogs, public dashboards, status pages.

dApp front-ends

Host a static front-end for an Ethereum dApp on Swarm. On-chain contracts with content-addressed UI for a fully decentralised stack.

What Swarm is *not*

- **Not a database.** Opaque blobs addressed by hash — no queries, no indexing, no search. Model structure in the app layer.
- **Not permanent without payment.** Content lives only while a valid postage stamp covers its chunks.
- **Not instantly consistent.** Uploads propagate asynchronously. `deferred:true` (default) returns once data is pushed; `deferred:false` waits for read-back.
- **Not free to write.** Every upload needs a funded light node and a stamp paid in xBZZ. Reads have a small free tier on gateways.
- **Not a smart-contract platform.** It stores data, it does not execute code. Complements Ethereum, does not replace it.

Limitations & gotchas

- **Stamp batch expiry.** Past expiry, chunks become eligible for pruning — not deleted instantly, but can vanish. Monitor and top up.
- **Dynamic pricing.** Stamp cost tracks a price that can rise — top up so batches outlast your needs.
- **Feed writes are append-only.** Each index slot writes once; always resolve the current index before writing.
- **Feed slots hold ≤ 4 KB.** For larger payloads, upload as a file first, then write the reference into the feed.
- **The public gateway blocks unreviewed sites.** A fresh upload can show "Forbidden" — view it at `bzz.limo/bzz/<hash>/` or via your own node.
- **Capacity & lifetime are non-deterministic.** Both depend on chunking and dynamic pricing — monitor and top up.

START BUILDING ON SWARM

THREE WAYS IN · PICK YOUR PATH

NO INSTALL · IN THE BROWSER

Upload via Bepport

Buy a stamp with any token and upload files or a whole site through Bepport's hosted gateway nodes — nothing to run, no node of your own.

bepport.ethswarm.org

Good for a first upload & quick demos; run a node when you want your own infrastructure.

YOUR OWN NODE · LINUX / MACOS / WSL

Run a Bee light node

```
$ curl -s https://raw.githubusercontent.com/ethersphere/bee/master/install.sh | TAG=v2.8.1 bash
```

```
$ npm i -g @ethersphere/swarm-cli # Node ≥ 18
```

Fund the node ↓ then buy a stamp — full steps: `/bee/installation/quick-start`

No funds? `bee dev` runs a local node with free stamps: `./.../bee-dev-mode`

OPTIONAL · AI-ASSISTED

Guided setup with Claude Code

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

```
$ git clone https://github.com/ethersphere/swarm-quickstart-skills.git
$ cd swarm-quickstart-skills && claude
```

Type `/swarm` — guided node setup, stamps, upload, hosting. Stuck? `/swarm-troubleshoot`. Windows: `code.claude.com/docs`



Prefer written docs? →
docs.ethswarm.org

The official developer docs — concepts, install, stamps, guides. Full link map on the back of this card.



Fund your node →
fund.ethswarm.org

A node without funds won't run. Top it up on Gnosis from any EVM chain, any token — one transaction.



Deploy a website →
justdeploy.bzz.link

Guided deploy walkthrough — Browser · Desktop app · CLI. Mirror: justdeploy.eth.limo

► More on the back: developer URLs, examples, network tools, community ►

Where to go next

scan by category · all links stay updated online



SWARM DEVELOPER RESOURCES

docs.ethswarm.org/docs/develop/resources

One curated landing page linking every category below. Bookmark it — links stay updated online, so this printed sheet keeps working.



LEARN

Get Started

your first 30 minutes

- **What is Swarm** /concepts/what-is-swarm
- **Developer intro** /develop/introduction
- **Bee node quick-start** /bee/installation/quick-start
- **Full install guide** /bee/installation/getting-started
- **Fund your node** — any EVM chain/token fund.ethswarm.org
- **Swarm papers** papers.ethswarm.org

SCAN QR → GET-STARTED HUB



Key Concepts

how it actually works

- **DISC** /concepts/DISC
- **Incentives & BZZ** /concepts/incentives/overview
- **Postage stamps** /.../buy-a-stamp-batch
- **Feeds** /.../tools-and-features/feeds
- **Manifests** /.../manifests · **Chunk types** /.../chunk-types

SCAN QR → DEEP-DIVE ON CONCEPTS



Developer Guides

recipes for common builds

- **Upload & download** /develop/upload-and-download
- **Host a website** /develop/host-your-website
- **Files & directories** /develop/files
- **Routing for SPAs** /develop/routing
- **Dynamic content (feeds)** /develop/dynamic-content
- **Multi-author blog · Access control (ACT)**

SCAN QR → ALL DEVELOPER GUIDES



EXAMPLES

Publishing & Websites

static sites, blogs, file uploads

- **upload-react** ethersphere/examples-js
- **Simple Blog** ethersphere/examples
- **Basic Static Website** ethersphere/examples

SCAN QR → MORE PUBLISHING EXAMPLES



Streaming

early HLS-over-Swarm work

- **Stream React example** [Solar-Punk-Ltd](https://solar-punk-ltd.com)
- **Stream JS library** [swarm-stream-js](https://swarm-stream-js.com)
- **Reference: feeds** /develop/dynamic-content

SCAN QR → STREAMING REPO



Messaging & Chat

PSS & GSOC patterns

- **Chat React example** [Solar-Punk-Ltd](https://solar-punk-ltd.com)
- **Chat JS** [swarm-chat-js](https://swarm-chat-js.com)
- **PSS messaging** /.../tools-and-features/pss
- **GSOC messaging** /.../tools-and-features/gsoc

SCAN QR → CHAT REPO



TOOLS

SDKs & APIs

talk to Bee from your code

- **bee-js** bee-js.ethswarm.org
- **Bee HTTP API** docs.ethswarm.org/api
- **swarm-cli · Swarm MCP** ethersphere/swarm-mcp
- **Bee (Go source)** ethersphere/bee

SCAN QR → BEE-JS DOCS



Gateways & Deploy

ship & inspect

- **Beeport** beeport.ethswarm.org
- **Public gateway** gateway.ethswarm.org
- **View by hash** bzz.limo/bzz/<hash>/
- **Local test network** /.../starting-a-test-network
- **SwarmScan** swarmscan.io

SCAN QR → DEPLOY & GATEWAY



Community & Support

stuck? ask here first

- **Discord** discord.ethswarm.org
- **Docs home** docs.ethswarm.org
- **Bee FAQ** /docs/bee/bee-faq
- **GitHub org** github.com/ethersphere
- **Foundation blog** blog.ethswarm.org

SCAN QR → JOIN DISCORD

